

MECH1634- Whiteboard Writing Project

Emmanuel Muhiga

MECH1634

Tuesday, December 5, 2023

Contents

1. Description of Operations	1
2. Point Location Table	1
3. I/O Listing	1
4. Observations	2
5. Demonstration Video	2

1. Description of Operations

The program waits for the operator to press the pushbutton, diPB1. After the button is pressed, pilot light 1, doPL1, will turn on. The robot will start writing the name on the whiteboard and draws a smiley figure below the name and at the center of the whiteboard. The robot will then wait 4 seconds before it erases the name first then the smiley figure. After the robot returns to a ready position.

2. Point Location Table

Name	Description
pReady	Ready position above the whiteboard
pLetterM(Appr,1-5,andDept)	All targets/points to write the letter M
pLetterU(Appr,1-5,andDept)	All targets/points to write the letter U
pLetterI(Appr,1-6,andDept)	All targets/points to write the letter I
pLetterH(Appr,1-6,andDept)	All targets/points to write the letter H
pLetterG(Appr,10-80,andDept)	All targets/points to write the letter G
pLetterA	Writes letter A with offsets from letter G target 60
pErase(Appr,10-40,andDept)	Erases the characters on the whiteboard

3. I/O Listing

Input		Output	
Name	Description	Name	Description
diPB1	Push button 1 - Normally open, momentary	doPL1	Pilot light 1, green, LED

4. **Observations**

Overall, the program was successful. The robot can write the name “MUHIGA” with a smiley figure below the name using different instructions. The main challenge was to calculate the position of the characters on the whiteboard based off the UCS. This can be improved by using a third-party software that can write the characters showing their geometry. (Inventor or AutoCAD).

5. **Demonstration Video**

<https://youtube.com/shorts/GrojTWQytiw?feature=share>